TASK- 1

Before coming to centralization cons which I have encountered, I would like to give a brief mention about it. As the name says, centralization is that system in which control and decision authority are concentrated within the hands of one entity or of a small group of entities. Decentralization, on the other hand, distributes control and authority throughout a network of independent participants, or nodes.

As an example in the context of blockchain, popular cryptocurrencies like Bitcoin and Ethereum's Ether are decentralized. Unlike traditional currencies, which are controlled by central banks, decentralized cryptocurrencies are governed by their programming code. Their monetary policies are shaped collectively by the community, ensuring no single authority has control.

Let's consider some situations which I feel where centralization creates trouble:

1. Centralized systems are when one authority controls the data. These systems can face many risks. Because all data is stored in one place, a **breach** can reveal a lot of sensitive information. This can cause problems like **identity theft, financial fraud,** and **privacy violations**. Also, centralized control usually makes it hard to see what is happening. This can make it difficult for users to know how their data is shared or how breaches are managed. Centralized groups have control over what gets shared and how data moves, which can lead to bias or false information. And if that one main system stops working, then there is a problem.

   Decentralization solves these problems by spreading control among many groups, which lowers the chance of failure in one place. It encourages openness because actions and transactions are kept on **unchangeable public records**. Decentralized systems are also stronger because they keep working even if one part fails, which makes them safer and more available for users. This change improves user independence and trust, giving a safer and clearer choice than centralized control.
   Data of 750 Million Indian Mobile Subscribers Sold on Hacker Forums. The blog highlights that centralized systems can increase the risk of data breaches.

2. As far as I am concerned with my regular short-term UPI payment failures, this situation seems tempting to me. Let's assume, a person whose account turns out to be frozen suddenly over some minor error, something maybe on exceeding a certain amount in transactions or perhaps suspicious behavior that has been associated with fraud. This circumstance happens as they are

unable to get access to their cash up to the time an extensive study is done and finalized that may take several days up to weeks before reaching an end.

The **problem** concerned with this is that banks act as a central authority controlling access to funds, hence leaving the user dependent on them for making transactions.
Decentralization in **cryptocurrencies** like Bitcoin provides individuals with complete control over their finances without relying on intermediaries like banks. Here’s how this makes a difference. With decentralized cryptocurrencies, one owns the private keys to the wallet. These keys are like the password to access our funds. Unlike traditional bank accounts where banks have authority over your assets, cryptocurrencies allow us to manage and transfer our money directly without needing permission from a central entity.

3. In the bureaucratic system of government, decisions are made by state officials rather than by elected representatives. Its characteristic structure is hierarchical and features rules that are formalized in labor division. This may ensure efficiency and fairness but always causes overload with paperwork, slow decision making, and inflexibility.
   More in the sense of consistency and control, centralization brings its share of benefits but problems, too. Centralized decision-making slows up processes, stalls innovation, and demoralizes staff. Rigid rules limit flexibility, and multiple approvals delay decisions. Most importantly, leaders of a centralized type might be unfamiliar with the surroundings and this leads to inefficiency and wasteful use of resources. This may damage an organisation's capacity to change and succeed.
   In this case, **decentralization** would help address the problems of slow decision-making and inflexibility associated with centralization. By distributing decision-making powers, it would allow for quicker responses to issues as local managers or staff members would be able to make decisions without waiting for multiple approvals from higher-ups. This would reduce delays, speed up processes, and foster a more adaptable work environment.